Signals and Systems — Tutorial 10

System Identification

Tutorial information

- It is expected that you attempt the prework questions before coming to the tutorial each week and submit your *handwritten* solutions to your tutor within the first 15 minutes of your tutorial.
- Not all questions are equally difficult. Difficult questions are marked with a $\star$.
- Questions marked **Extension** go beyond the assessable content for this course and are provided to give a deeper theoretical background and understanding.
- You may not be able to complete all questions within the tutorial time. It is *strongly* recommended that you attempt all questions in your own time and ask your tutor if you are stuck.

Prework questions

A: Linear System Response

Recall the response of a linear system with transfer function $G(s)$ to an input $u(t) = A\cos(\omega_0 t)$ is

$$y(t) = A|G(j\omega_0)|\cos\left(\omega_0 t + \angle G(j\omega_0)\right)$$

The responses $y_b(t), y_r(t)$ and $y_g(t)$ of an unknown system $y(t) = f(u(t))$ to the inputs $u_b(t) = \cos(t), u_r(t) = \cos(2t)$ and $u_g(t) = \cos(4t)$ are shown in Figure 1. Determine whether the system could be linear.

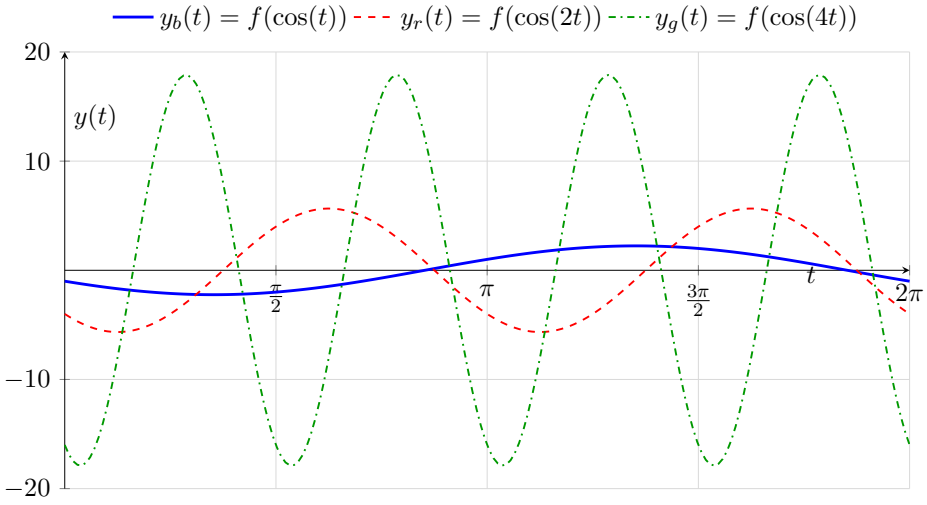


Figure 1: Output of the unknown system to the inputs $u_b(t), u_r(t)$ and $u_g(t)$.

B: Frequency Response

Suppose we test a system with a pulse-train input $u(t)$ with period $T = 1\,\text{s}$ and pulse width $\tau = 0.5\,\text{s}$ shown in Figure 2.

1. Use the Fourier series for $u(t)$ to determine the frequencies at which the system response has been tested.

2. As we increase the period T do we test more or fewer frequencies? Explain.

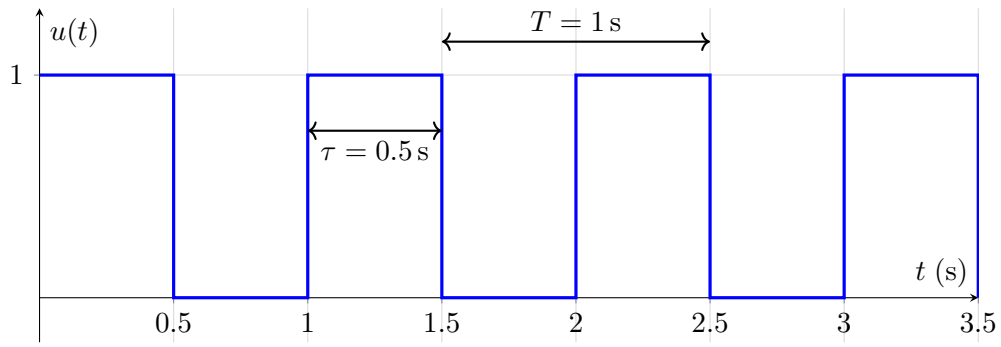


Figure 2: Pulse-train with period $T = 1\,\text{s}$ and duration $\tau = 0.5\,\text{s}$.

1: Follow-up Measurements

The response of the unknown system $y = f(u(t))$ from the prework to the superposition of $u_b = \cos(t)$ and $u_r(t) = \cos(2t)$ is measured and is shown in Figure 3. Based on this additional measurement do you consider the system to be linear or nonlinear?

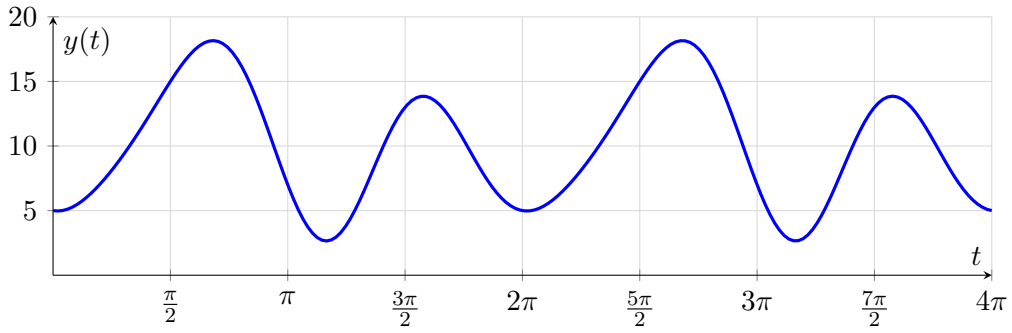


Figure 3: Response of $y(t) = f(u(t))$ to $u(t) = \cos(t) + \cos(2t)$.

2: System parameters from Bode plots

1. Given the Bode diagram shown in Figure 4 determine the DC gain (K), damping ratio (ζ), percentage overshoot (M_p), natural frequency (ω_n), damped oscillation frequency (ω_d), rise-time (t_r) and settling time (t_s) of the system.
2. Sketch the step-response of the system indicating the rise-time and settling time on the sketch.
3. Suppose ω_n does not change but the system becomes critically damped. Sketch the step response of the system.

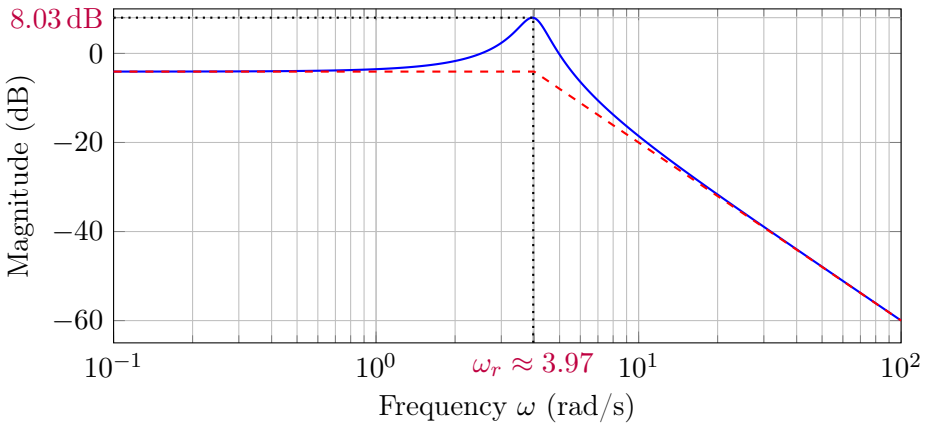


Figure 4: Bode Diagram for an unknown system

3: Step responses and zeros

In order to identify the dynamics of three systems with transfer functions $G_b(s), G_g(s)$ and $G_r(s)$ the responses $y_b(t), y_g(t)$ and $y_r(t)$ to a step input to each system are measured and shown in blue, green and red respectively, in Figure 5.

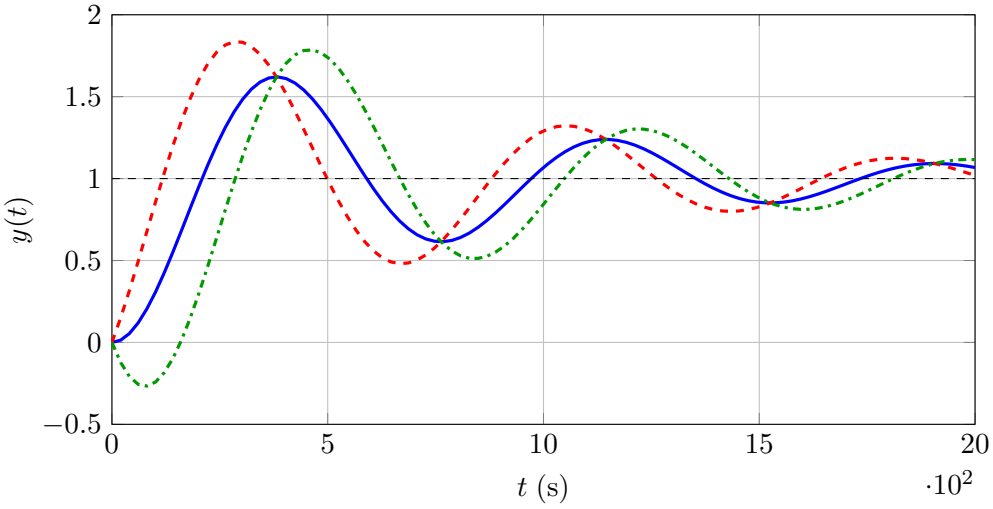


Figure 5: Step responses of three unknown second-order systems

The maximum values of each response are $\max\{y_b(t)\} = 1.62, \max\{y_g(t)\} = 1.84$ and $\max\{y_r(t)\} = 1.79$ and the initial slopes for each response are $\dot{y}_b(0) = 0, \dot{y}_g(0) = 7 \times 10^{-3}$ and $\dot{y}_r(0) = -7 \times 10^{-3}$. The period for all three responses is $T = 762.1\,\text{s}$.

1. Assume that $y_b(t)$ has no zeros. For the other two systems identify whether the zeros are RHP or LHP zeros.
2. Assuming all the systems are second order find the DC gain, damping ratio and natural frequency of each system.
3. Determine the location of the zero of each system and give the transfer function for each system.
4. Could the transfer functions all have the same pole locations?

4: Transients

Suppose we model a system with the transfer function

$$G(s) = \frac{10}{s^2 + 2s + 4}$$

To test the model we apply an input $u(t)$ and measure the output $y(t)$.

1. Sketch the asymptotic Bode magnitude and phase plot for the system model.
2. Find the magnitude and phase of the response of the system to the input $u(t) = \sin(2t)$.
3. How long after $t = 0$ should we wait until measuring the response of the system to ensure the data we collect is within 2% of the steady-state response of the system?

5: System Identification from Noisy Data

An unknown stable continuous-time system is assumed to have the model

$$G(s) = \frac{K}{\tau s + 1}.$$

A step input of amplitude $A = 2.5$ is applied at $t = 0$. The measured output is

t (s)	0	0.5	1	2	3	4	6	10
$y(t)$	0.02	0.91	1.53	2.41	2.89	3.18	3.48	3.59

1. Estimate the steady-state output $y(\infty)$, DC gain K and time constant τ.
2. Derive the complete predicted step response $y(t)$ and therefore the impulse response $h(t)$ of the unknown system.
3. Given that the system is subject to noise is this an appropriate method to find the system's impulse response. How would this compare to identifying the parameters of the transfer function by measuring the frequency response of the circuit?

6: Second-Order Identification

A system is assumed to have a transfer function of the form

$$G(s) = \frac{K\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}. \tag{1}$$

A step input $1.5H(t)$ produces a response with

$$y(\infty) = 3.00, \qquad y_{\text{pk}} = 4.05 \quad \text{and} \quad t_p = 0.80 \text{ s}$$

where y_{pk} is the first peak value and t_p is the time at which the first peak occurs.

1. Determine the DC gain K and percentage overshoot M_p.
2. Determine the damping ratio ζ, natural frequency ω_n and find the poles of the transfer function (1).
3. Sketch the step response, indicating the rise-time and settling time on the sketch.
4. For the value of ζ found above is the step response a valid technique to identify the parameters of the system?
5. **(Extension)** Let $M_p' = M_p/100$. From the step response of (1), prove that

$$M_p' = \exp\left(-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}\right).$$

and hence show that

$$\zeta = \frac{-\ln(M_p')}{\sqrt{\pi^2 + \left(\ln(M_p')\right)^2}}$$